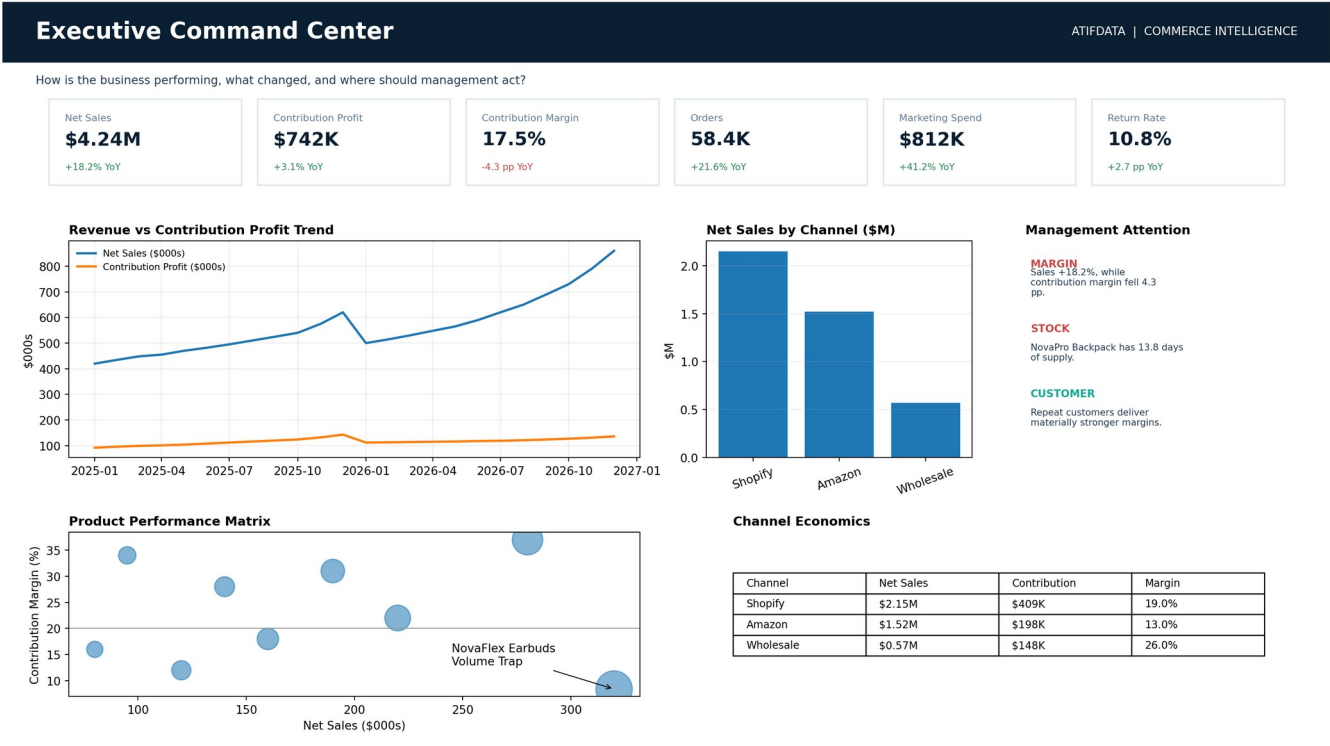


Commerce Intelligence

Omnichannel Retail Data Modernization & Analytics

Solution Architecture • Data Model • Business Rules • Executive Analytics



AtifData | Data Modernization & Analytics Solutions

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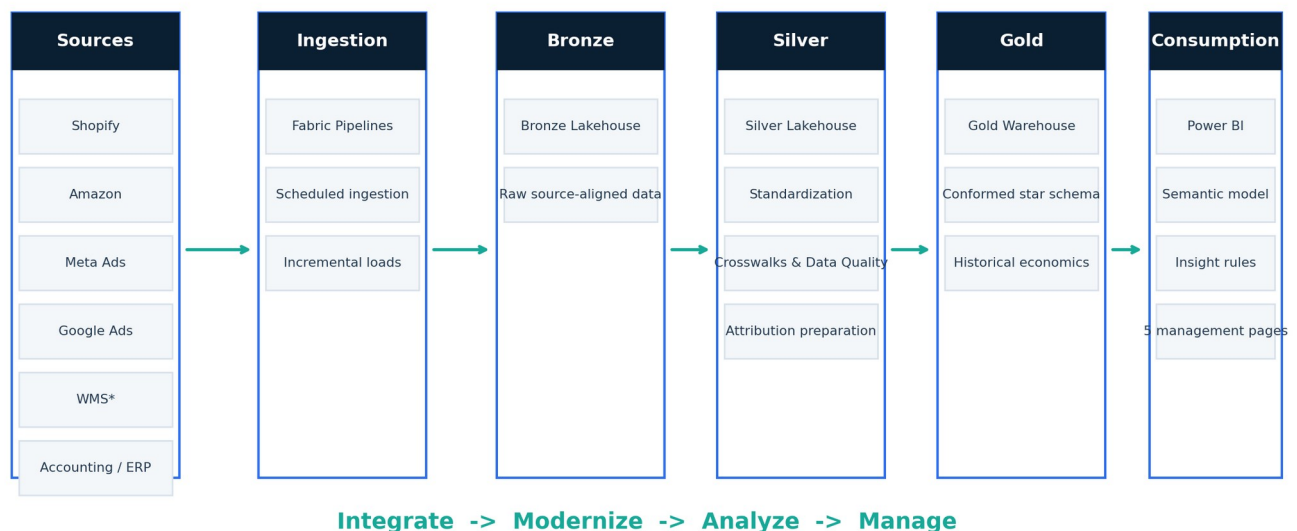
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1. Executive Summary

Commerce Intelligence is an end-to-end data modernization and analytics solution for omnichannel retail - a retail operating model in which sales and customer interactions span multiple connected channels such as an ecommerce website, online marketplaces and wholesale channels. The solution unifies fragmented sales, marketing, customer, inventory and financial data into a governed analytical model for management decision-making.

The architecture combines Microsoft Fabric, layered Bronze/Silver/Gold processing, a conformed multi-fact star schema, a Power BI semantic model and an executive insight layer. The objective is to move beyond isolated platform reporting and show where economic value is being created or lost.

Commerce Intelligence - End-to-End Architecture



* WMS = Warehouse Management System

2. Business Context and Management Challenges

Nova Retail Group operates across Shopify, Amazon and wholesale channels while acquiring customers through Meta and Google advertising. Inventory is managed through a Warehouse Management System (WMS) across multiple locations, while product costs and operating expenses are maintained separately in accounting systems.

Management can see sales in individual platforms, but cross-functional questions are harder: Which products remain profitable after returns and acquisition costs? Which marketing channels acquire valuable customers? Which high-margin products are at risk of stockout? Are rising revenues translating into stronger contribution profit?

3. Solution Objectives

Create a single, reliable analytical foundation across commerce, marketing, inventory and finance; standardize inconsistent identifiers and business definitions; preserve historical product economics; support auditable attribution and profitability logic; deliver management-ready Power BI analytics; and surface actionable signals rather than isolated KPIs.

4. Source Systems

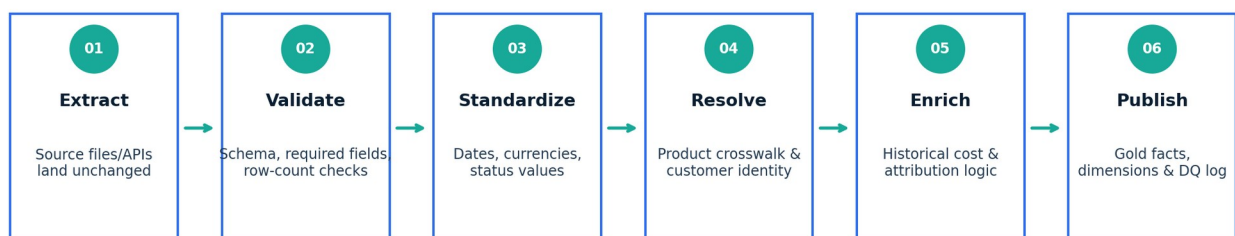
Shopify provides orders, order lines, customers, products, discounts, refunds and attribution fields. Amazon provides orders, items, marketplace fees, returns and ASIN/SKU mappings. Meta Ads and Google Ads provide daily campaign spend, impressions, clicks, conversions and platform-attributed revenue. The Warehouse Management System (WMS) provides daily inventory snapshots and movements. Accounting/ERP provides effective-dated product costs and monthly operating expenses.

5. Data Quality, Crosswalks and Standardization

A crosswalk is a mapping table used to reconcile different identifiers for the same business entity across source systems. For example, Shopify SKU NRG-WB-001, Amazon identifier B0NRG1024, WMS code WB001 and ERP code PROD-WB01 can all resolve to one governed MasterSKU. Data Quality (DQ) controls cover deduplication, missing required fields, invalid dates, unmapped products, status standardization, currency conversion, late-arriving returns and source-to-target reconciliation.

Silver-layer processing standardizes these inconsistencies before publishing governed analytical entities to Gold. Operational controls capture rows received, accepted, rejected, duplicated, unmapped and reconciled.

Fabric Pipeline & Data Quality Processing Flow



Data Quality controls: received • accepted • rejected • duplicates • unmapped SKUs • invalid dates • reconciliation variance

Product Crosswalk - Source-to-Master Resolution

Source	Source Identifier	Silver Mastering	Gold Target
Shopify	NRG-WB-001	MasterSKU NRG-WB-001	DimProduct.ProductKey 1042
Amazon	B0NRG1024 / AMZ-WB01	MasterSKU NRG-WB-001	DimProduct.ProductKey 1042
WMS	WB001	MasterSKU NRG-WB-001	DimProduct.ProductKey 1042
ERP	PROD-WB01	MasterSKU NRG-WB-001	DimProduct.ProductKey 1042

A crosswalk maps different source-system identifiers for the same business entity to one governed master identifier.

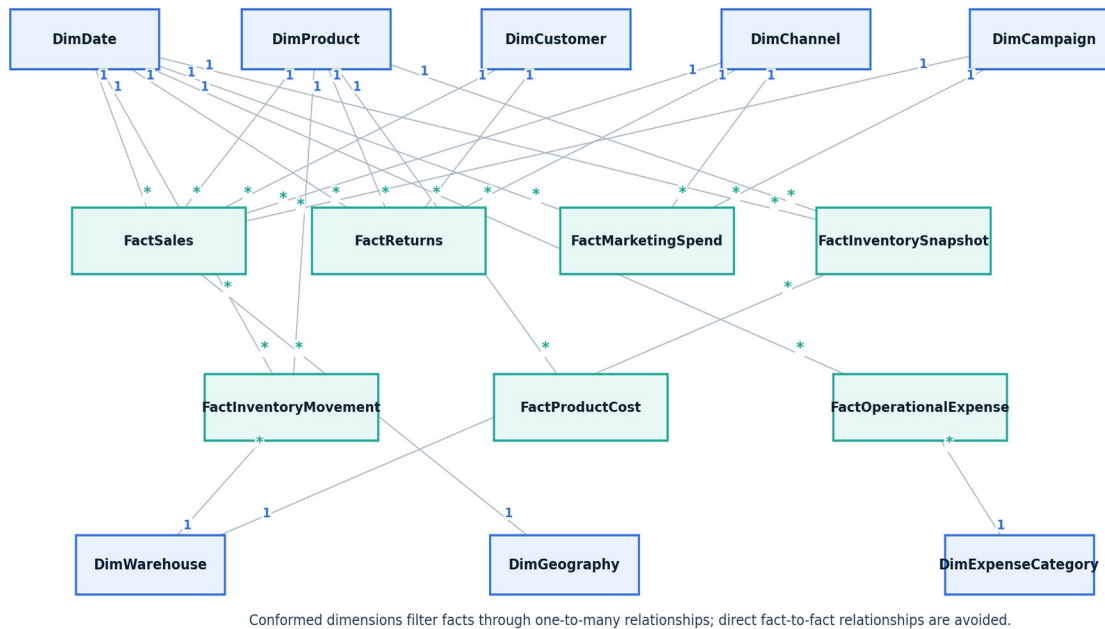
6. Gold Analytical Model

The Gold layer uses conformed dimensions with multiple facts at explicit grains. Core dimensions are Date, Product, Customer, Channel, Campaign, Warehouse and Geography, with Expense Category where required. Core facts are Sales, Returns, Marketing Spend, Inventory Snapshot, Inventory Movement, Product Cost and Operational Expense.

FactSales is one row per order line. FactReturns records return events on their actual return date. FactMarketingSpend is campaign-platform-day. FactInventorySnapshot is SKU-warehouse-day. Product cost is effective-dated so historical COGS is not recalculated using current costs.

Gold Layer - Conformed Multi-Fact Star Schema

Cardinality: 1 = dimension side, * = fact side



7. Business Rules and Profitability

Gross Sales equals quantity multiplied by original unit price. Net Revenue deducts discounts and refunds. COGS uses the unit cost effective on the transaction date. Contribution Profit deducts COGS, marketplace fees, payment fees, fulfillment costs and allocated marketing cost from Net Revenue. General operating expenses remain separate for company-level operating profitability.

Marketing attribution uses a 30-day last non-direct-touch rule for the core analytical view while retaining platform-reported attribution for comparison. Product-level marketing allocation is proportional to attributed net sales where campaigns map to products or categories.

8. Power BI Experience

The reporting experience is organized into five decision-focused pages: Executive Command Center, Product & Profit Intelligence, Customer Intelligence, Marketing Intelligence and Inventory Intelligence. Each page answers a specific management question and supports navigation from enterprise-level signals to underlying commercial drivers.

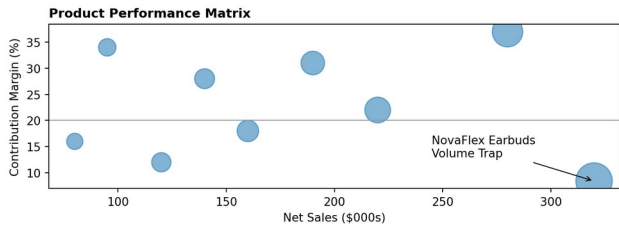
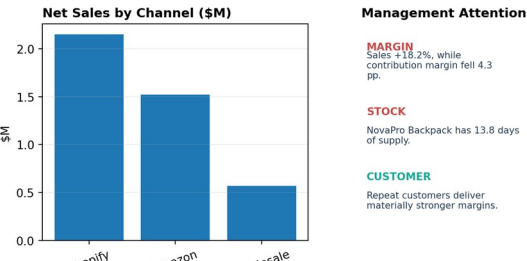
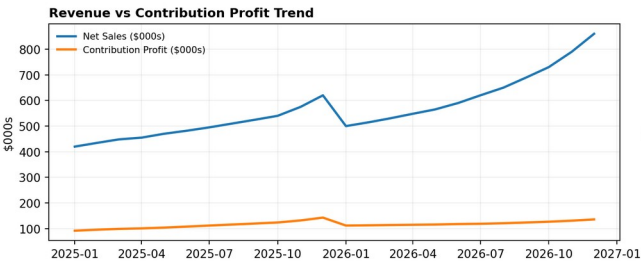
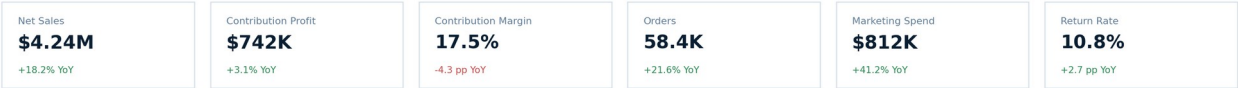
In the acquisition cohort visual, M0 represents the customer's acquisition or first-purchase month; M1 is one month after first purchase, continuing through M5 at five months after first purchase. Rows represent acquisition cohorts, while columns represent elapsed months since first purchase.

Executive Command Center

Executive Command Center

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How is the business performing, what changed, and where should management act?



Channel Economics

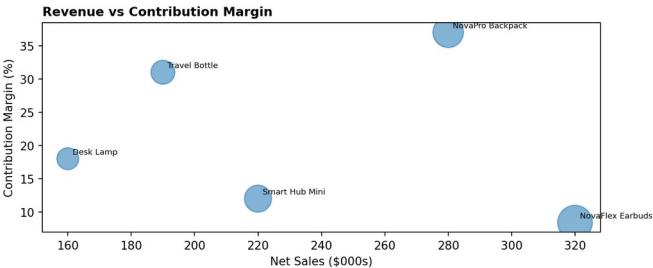
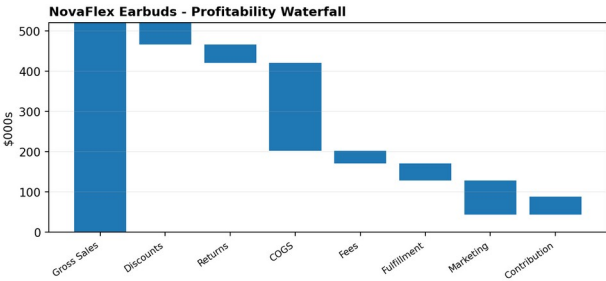
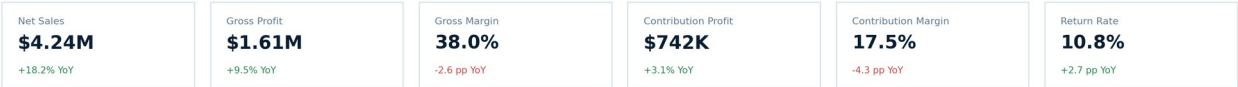
Channel	Net Sales	Contribution	Margin
Shopify	\$2.15M	\$409K	19.0%
Amazon	\$1.52M	\$198K	13.0%
Wholesale	\$0.57M	\$148K	26.0%

Product & Profit Intelligence

Product & Profit Intelligence

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Which products are actually creating or destroying economic value?



Product Profitability Ranking

Product	Net Sales	Growth	Gross Margin	Contribution Margin	Return Rate
NovaFlex Earbuds	\$320K	+29%	31.2%	8.4%	14.8%
NovaPro Backpack	\$280K	+24%	48.5%	37.0%	4.6%
Smart Hub Mini	\$220K	+20%	34.0%	12.0%	17.1%
Travel Bottle	\$190K	+12%	44.2%	31.0%	5.1%
Desk Lamp	\$160K	+8%	37.0%	18.0%	8.3%

Customer Intelligence

Customer Intelligence

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Who are our valuable customers, where did they come from, and are they returning?

Customers 41.2K +16.4% YoY	New Customers 24.8K +22.1% YoY	Repeat Customers 16.4K +9.0% YoY	Repeat Purchase Rate 39.8% +3.4 pp YoY	AOV \$82.40 +5.2% YoY	180-Day Value \$111 +12.7% YoY
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New vs Repeat Customer Economics

Metric	New	Repeat	Difference
AOV	\$72	\$94	+31%
Contribution Margin	17%	29%	+12 pp
Return Rate	11%	6%	-5 pp
Discount Rate	14%	7%	-7 pp

AOV is in USD; rate metrics are percentages. pp = percentage points.

Acquisition Cohort - Repeat Activity Index

Acquisition Cohort	M0	M1	M2	M3	M4	M5
Jan	100	31	24	20	18	16
Feb	100	33	25	22	19	17
Mar	100	35	27	23	20	18
Apr	100	36	28	24	21	19
May	100	38	29	25	22	20
Jun	100	39	31	27	23	21

M0 = acquisition month; M1 = one month later; ...; M5 = five months later

Acquisition Channel Quality

Acquisition Source	CAC	First Order	90D Value	180D Value	Repeat Rate
Meta	\$32	\$78	\$82	\$88	22%
Google	\$41	\$70	\$101	\$128	37%
Organic / Direct	-	\$84	\$109	\$136	42%
Other	\$36	\$73	\$91	\$106	29%

Marketing Intelligence

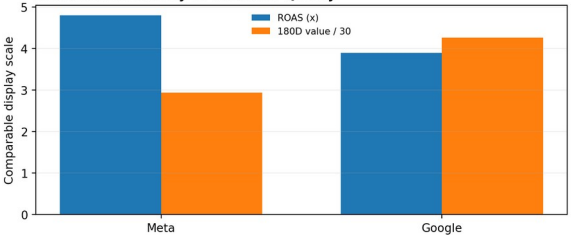
Marketing Intelligence

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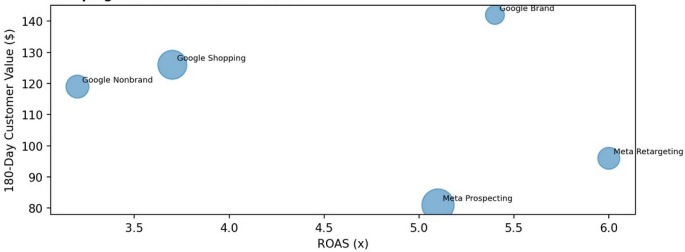
Are we buying profitable growth, or simply buying revenue?

Ad Spend \$812K +41.2% YoY	Attributed Revenue \$3.48M +26.0% YoY	ROAS 4.29x -0.52x YoY	New Customers 24.8K +22.1% YoY	CAC \$36.50 +15.4% YoY	Contribution ROAS 1.31x -0.18x YoY
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Short-Term Efficiency vs Customer Quality



Campaign ROAS vs Customer Value



Campaign Profitability & Attribution

Campaign	Spend	Platform Revenue	Unified Revenue	ROAS	CAC	Contribution ROAS
Meta Prospecting	\$260K	\$1.33M	\$1.19M	4.58x	\$34	1.22x
Meta Retargeting	\$120K	\$720K	\$684K	5.70x	\$27	1.48x
Google Brand	\$90K	\$486K	\$454K	5.04x	\$29	1.67x
Google Shopping	\$210K	\$777K	\$735K	3.50x	\$43	1.25x
Google Nonbrand	\$132K	\$422K	\$417K	3.16x	\$46	1.08x

Inventory Intelligence

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Do we have the right products, in the right quantities, in the right locations?

Inventory Value

\$1.18M

+8.2% YoY

Available Units

31.6K

-4.7% YoY

Inventory Turnover

5.8x

+0.4x YoY

Stockout Risk

7 SKUs

+3 SKUs

Slow-Moving Value

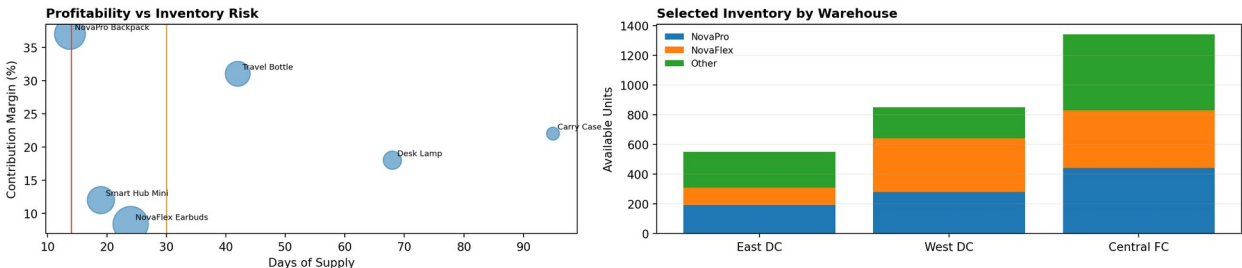
\$186K

+11.0% YoY

Revenue at Risk

\$214K

Current estimate



Inventory Attention List					
Product	Available	30D Units	Days Supply	Contribution Margin	Risk
NovaPro Backpack	550	1,195	13.8	37.0%	Critical
Smart Hub Mini	590	930	19.0	12.0%	Watch
NovaFlex Earbuds	1,248	1,560	24.0	8.4%	Watch
Travel Bottle	1,092	780	42.0	31.0%	Healthy
Carry Case	665	210	95.0	22.0%	Slow-moving

9. Business Impact & Management Outcomes

The integrated analytical view surfaces quantified management outcomes across growth, profitability, customer quality and inventory. Net Sales increased 18.2%, while Contribution Margin declined 4.3 percentage points, exposing a gap between top-line growth and economic value creation. NovaPro Travel Backpack delivered a 37.0% contribution margin but had only 13.8 days of supply, highlighting a replenishment priority.

Google-acquired customers showed \$128 of 180-day value versus \$88 for Meta-acquired customers - approximately 45% higher - despite Meta's stronger short-term ROAS. Smart Hub Mini reached a 17.1% return rate, exposing a concentrated product issue. Repeat customers delivered a 29% contribution margin versus 17% for new customers, a 12 percentage-point advantage that supports greater focus on retention economics.

Business Impact & Management Outcomes



10. Executive Insight Layer

Semantic measures feed deterministic insight rules. Examples include revenue growth above 10% combined with contribution-margin deterioration greater than two percentage points; high-margin products with fewer than 14 days of supply; and cases where short-term channel ROAS conflicts with 180-day customer value. This creates an auditable KPI-to-insight-to-action pattern.

11. Implementation Approach

1. Ingest source-aligned data through Fabric pipelines into Bronze. 2. Validate, standardize and reconcile in Silver. 3. Resolve product identifiers and customer references. 4. Apply historical cost and attribution logic. 5. Publish conformed Gold facts and dimensions. 6. Build the Power BI semantic model and measures. 7. Configure insight rules and management pages. 8. Operationalize refresh monitoring, data-quality controls and enhancement processes.

12. Technology Stack

Microsoft Fabric Data Factory / Pipelines; Fabric Lakehouse and Warehouse; SQL and Python for transformation and validation where appropriate; Power BI semantic modeling and DAX; governed dimensional modeling; scheduled data-quality and reconciliation controls.

Appendix A. KPI Catalogue

Domain	Core KPIs
Executive / Sales	Gross Sales; Net Revenue; Orders; Units Sold; AOV; Revenue Growth; Units per Order
Profitability	COGS; Gross Profit; Gross Margin %; Contribution Profit; Contribution Margin %; Operating Profit
Customer	New Customers; Repeat Customers; Repeat Purchase Rate; Purchase Frequency; 90-Day Value; 180-Day Value; CAC; LTV:CAC
Marketing	Spend; Impressions; Clicks; CTR; CPC; Conversion Rate; ROAS; CAC; Contribution ROAS
Product	Net Sales; Units; ASP; Gross Margin; Contribution Margin; Return Rate; Sales Growth; Profit Growth
Inventory	Stock on Hand; Available Units; Inventory Value; Sales Velocity; Days of Supply; Inventory Turnover; Stockout Risk; Slow-Moving Value

Appendix B. Source-to-Target Summary

Source	Primary Gold Targets
Shopify	FactSales, FactReturns, DimCustomer, DimProduct, DimChannel, DimCampaign
Amazon	FactSales, FactReturns, DimProduct, DimChannel
Meta Ads	FactMarketingSpend, DimCampaign
Google Ads	FactMarketingSpend, DimCampaign
WMS	FactInventorySnapshot, FactInventoryMovement, DimWarehouse
Accounting / ERP	FactProductCost, FactOperationalExpense, DimExpenseCategory

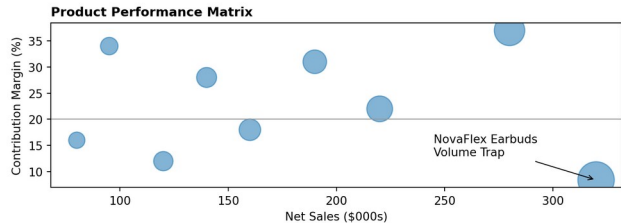
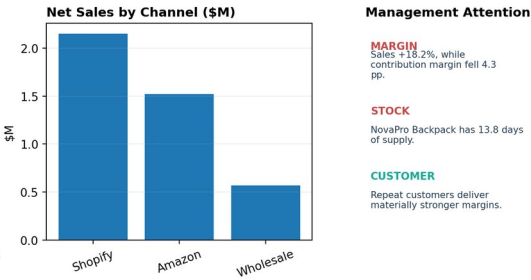
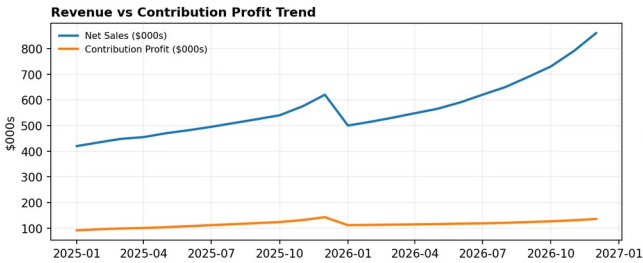
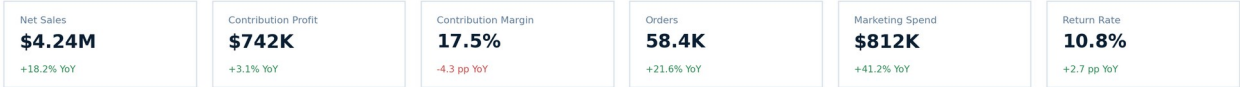
Appendix C. Power BI Page Gallery

Executive Command Center

Executive Command Center

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Channel Economics

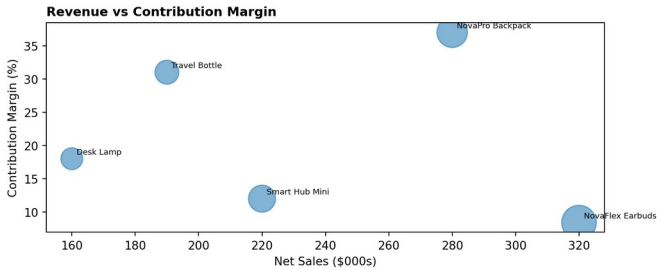
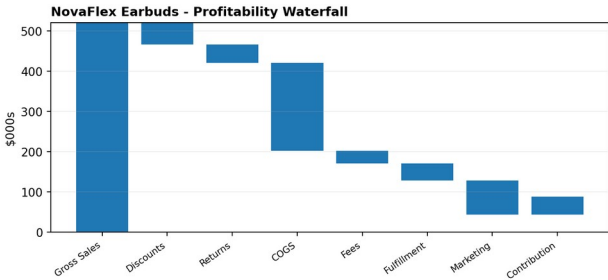
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Product & Profit Intelligence

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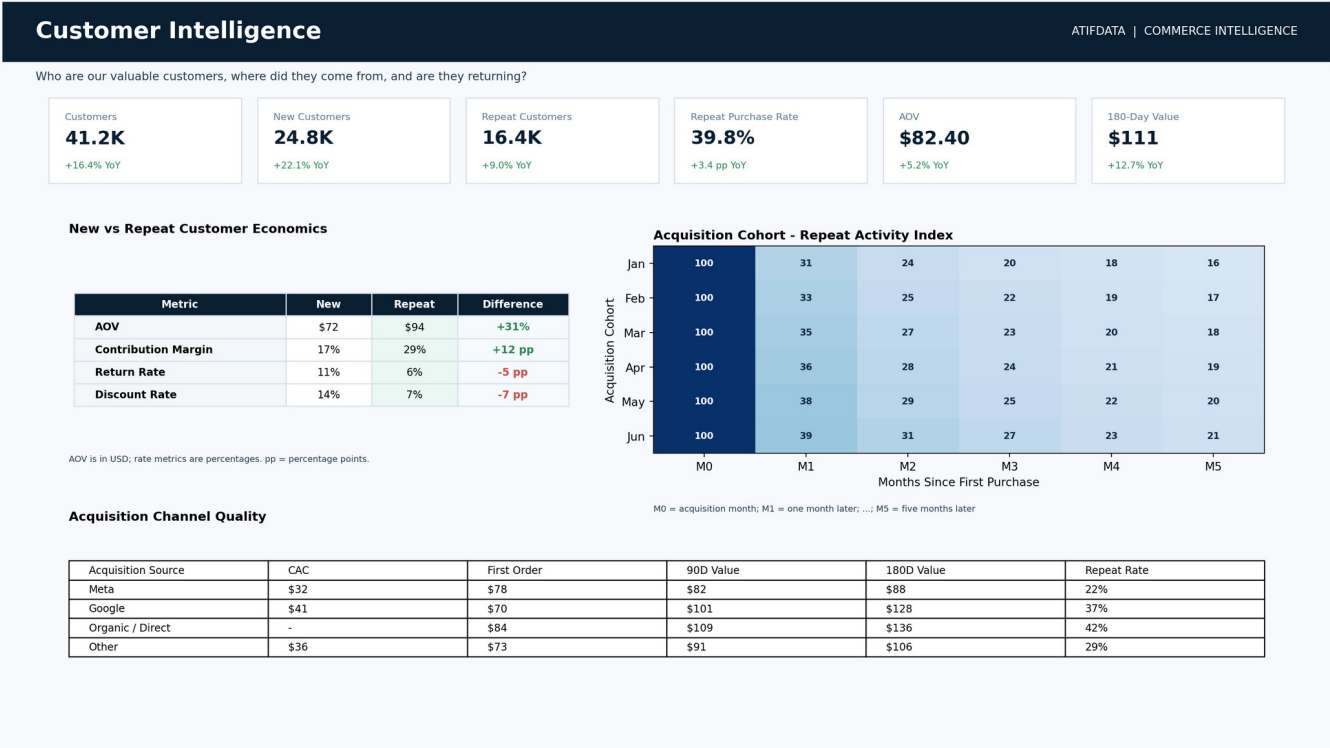
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Customer Intelligence



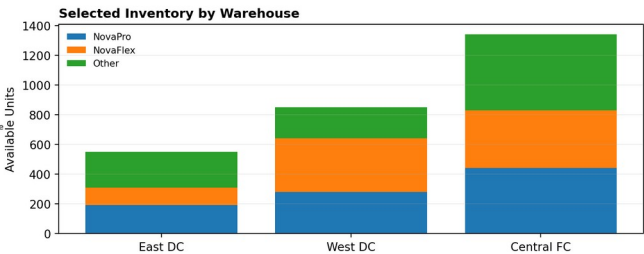
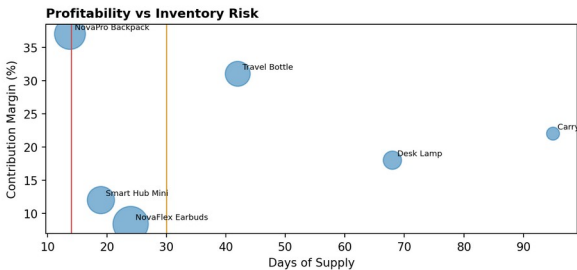
Inventory Intelligence

Inventory Intelligence

ATIFDATA | COMMERCE INTELLIGENCE

Do we have the right products, in the right quantities, in the right locations?

Inventory Value \$1.18M +8.2% YoY	Available Units 31.6K -4.7% YoY	Inventory Turnover 5.8x +0.4x YoY	Stockout Risk 7 SKUs +3 SKUs	Slow-Moving Value \$186K +11.0% YoY	Revenue at Risk \$214K Current estimate
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Inventory Attention List

Product	Available	30D Units	Days Supply	Contribution Margin	Risk
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